

Agernesh Dereje

Full Stack Developer

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Professional Summary

Backend-focused Full Stack Developer with a passion for building robust APIs and efficient database architectures. Experienced in full software development lifecycles, from design to deployment, with a focus on solving real-world challenges through technology. Proven ability to collaborate in agile environments and deliver production-ready systems for healthcare, education, and agriculture.

Education

B.Sc. in Information Technology Bahir Dar Institute of Technology (BiT)
September 2023 – June 2026

- ✓ Relevant Coursework: Data Structures & Algorithms, Database Design, Project Management, Network Design.

Professional Experience

Kirsoft Software Consulting, Lettering System

- ✓ Engineered a comprehensive lettering system using PHP, JavaScript, and MySQL, featuring unique ID generation and automated email notifications.
- ✓ Managed development sprints using Jira/Trello and maintained version control with Git/GitHub in an agile environment.
- ✓ Delivered a production-ready system utilized by multiple organizations, enhancing operational efficiency and communication workflows.

Key Projects

Agricultural Input Budget Optimization Model , AI Developer

- ✓ Developed a predictive AI model utilizing Linear and Random Forest regression to optimize agricultural input allocation based on granular soil and climate data.
- ✓ Designed a greedy optimization algorithm to maximize farmer ROI while strictly adhering to budget constraints.
- ✓ Built a dual-interface system featuring a web-based dashboard and mobile SMS integration to ensure accessibility for end-users.

National Health Management System , Lead Developer (<https://ager-indol.vercel.app/>)

- ✓ Engineered a high-concurrency, full-stack platform (React, Node.js, PostgreSQL) to support hierarchical administrative structures across federal and regional levels.
- ✓ Implemented enterprise-grade Role-Based Access Control (RBAC), real-time analytics, and proactive error monitoring to ensure system reliability.
- ✓ Multi-Factor Authentication (MFA): Implemented TOTP-based 2FA using authenticator apps, with backup codes for account recovery

- ✓ Single Sign-On (SSO): Integrated OAuth 2.0 and SAML 2.0 protocols for seamless enterprise authentication
- ✓ JWT-Based Authentication: Secure token generation with short-lived access tokens (15 min) and refresh tokens (7 days) stored in HTTP-only cookies
- ✓ Session Management: Implemented Redis-based session storage with automatic invalidation on role changes or suspicious activity
- ✓ Password Policy Enforcement: Mandatory complex password requirements (minimum 6 characters, special characters, numbers) with bcrypt hashing (salt rounds: 6)
- ✓ Account Lockout Mechanism
- ✓ Email/OTP Verification: Email verification for new accounts and OTP-based password reset flows
- ✓ **Awarded "Best Project" by Bahir Dar University Computing Faculty**

Student Management System ,Full Stack Developer

- ✓ Developed a React and MySQL-based system for remote learning, featuring secure authentication and Q&A forums.
- ✓ Optimized database performance through normalized schema design and strategic indexing to support academic progress tracking.

Technical Skills

- ✓ **Frontend:** React.js, Next.js, JavaScript (ES6+), Tailwind CSS, HTML5, CSS3, Bootstrap
- ✓ **Backend & Databases:** Node.js, Express.js, PHP, PostgreSQL, MySQL, MongoDB
- ✓ **Tools & DevOps:** Git, GitHub, Vercel, Jira, Trello, ClickUp, JWT Authentication
- ✓ **Languages & Competencies:** Amharic (Native), English (Fluent); Agile Methodology,
- ✓ Strategic Leadership, Complex Problem Solving